TEAM FOREVERKNIGHTS

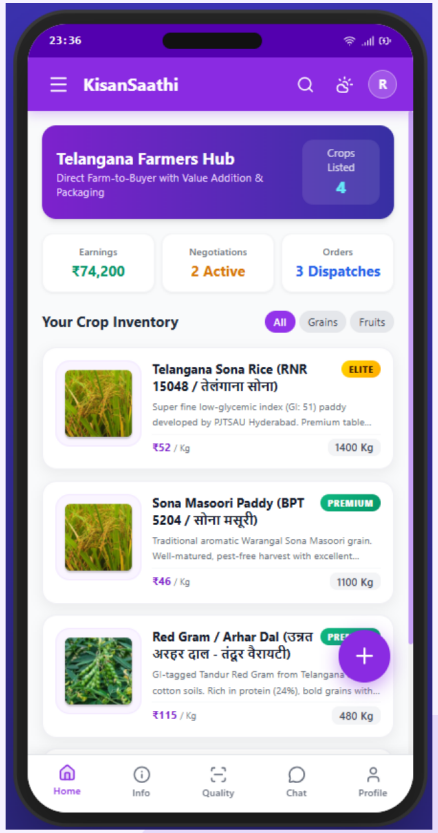
HACK-X 2026

TITLE PAGE

HACK-X 2026
35-Hour Hackathon · MJCET

KisanSaathi (किसान साथी)

- **Tagline –** Empowering Farmers with Direct Mandi Access, Computer Vision Quality Grading & Cluster Logistics
- **Theme –** Agriculture, FoodTech & Rural Development
- **Problem Statement –** Elimination of intermediaries, subjective mandi quality exploitation, and prohibitive freight costs for smallholder farmers
- **Team Name –** Team ForeverKnights
- **Team Members –** Gulam Waheed Mujtaba – Lead Architect & Full Stack Lead Shaik Mohammed Ateeq – Cloud Infrastructure & AI Systems Engineer Malik Samiullah Khan – UI/UX & Agricultural Tech Specialist
- **Live Prototype –** PWA with bilingual AI assistant, Leaflet GPS route tracking and escrow wallet



KisanSaathi – Farmers Hub

TEAM FOREVERKNIGHTS

PROPOSED SOLUTION

HACK-X 2026
35-Hour Hackathon · MJCET

1 · Instant Crop Listing
Upload produce details or scan a photo sample

2 · Automated AI Grading
Computer Vision predicts Elite / Premium / Classic and a fair price range

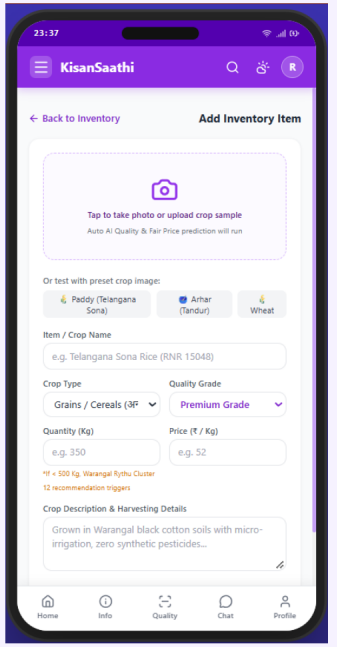
3 · Cluster Aggregation
Sub-500kg harvests pooled (Warangal Agri-Cluster #12), freight down up to 38%

4 · Direct Negotiation
Counter-offer desk with bulk buyers in Hyderabad & Secunderabad

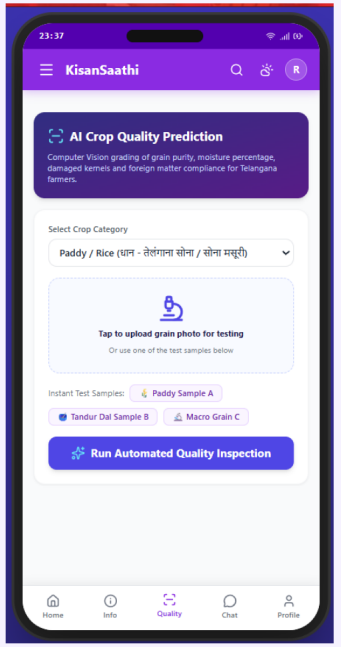
KisanSaathi
5-Step Workflow

5 · Escrow Settlement
DBT payout within 24 hours of delivery, with real-time GPS tracking

	Traditional Mandi	KisanSaathi
Commission	12–18% lost to middlemen	0% (direct escrow)
Quality	Subjective trader bias	Computer Vision grading
Freight	Costly individual transport	Cluster pooling, 38% savings
Payment	15–45 day credit cycle	24-hour DBT transfer



Step 1: crop listing



Step 2: AI grading

TEAM FOREVERKNIGHTS

TECHNICAL APPROACH

HACK-X 2026
35-Hour Hackathon · MJCET

System Architecture & Tech Stack

Frontend
Installable PWA for mobile and desktop
HTML5
Tailwind CSS

→

Backend
Node.js REST API, zero-dependency server
Node.js

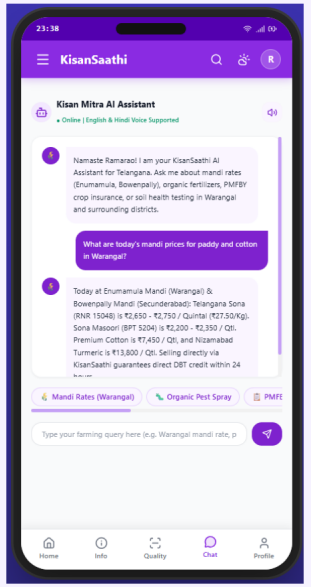
→

AI & Vision
Client-side defect ratio, grading and price
OpenCV
TF.js

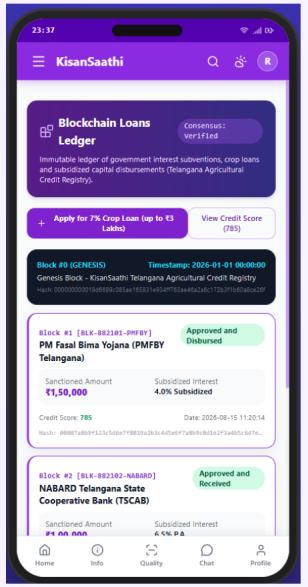
→

Assistant & Maps
English/Hindi Kisan Mitra, live Telangana route
Web Speech
Leaflet.js

Key innovations: offline-capable PWA · blockchain credit score (785/900) · 4% subsidized loan ledger



Bilingual AI assistant



Blockchain loan ledger

Implementation Methodology

1. Farmer input: photo or crop data
2. AI quality engine (OpenCV / TF.js)
3. Smart contract ledger for loans
4. Real-time negotiation marketplace
5. Logistics and GPS transit (Leaflet)

TEAM FOREVERKNIGHTS

FEASIBILITY AND VIABILITY

HACK-X 2026
35-Hour Hackathon · MJCET

Feasibility of the Idea

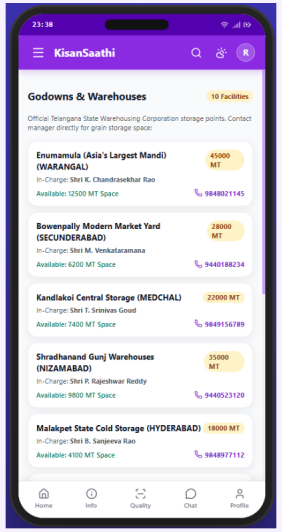
Technical
Client-side models, lightweight web stack, low bandwidth for rural 4G

Economic
Zero listing fee for farmers; 1.5% fee on bulk corporate deals

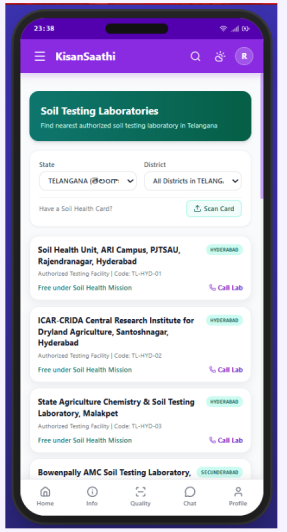
Operational
Pre-integrated with State Warehousing godowns and soil labs

Key Challenges → Mitigation Strategies

Digital literacy and language	→	**Bilingual voice assistant** Hindi & English with single-tap OTP login
Harvests under 500 Kg	→	**Cluster Aggregation Engine** Pools local farmers to command bulk pricing
Trust in e-payments	→	**Verified escrow** Blockchain-logged, tamper-proof ledger



Godowns directory



Soil labs directory

Built this hackathon
Bilingual prototype, AI crop grader, escrow wallet, lab directory

Next phase
Government e-NAM API integration and satellite NDVI

TEAM FOREVERKNIGHTS

IMPACT AND BENEFITS

HACK-X 2026
35-Hour Hackathon · MJCET

Smallholder Farmers →
- **20–28%** higher net income
- **Fair grading** no arbitrary quality deductions
- **24h payout** direct DBT via escrow

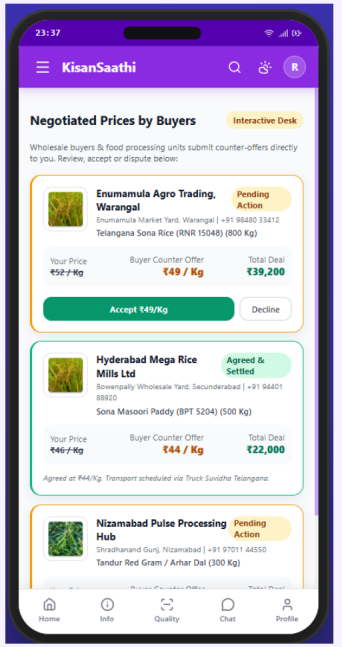
Buyers & Millers →
- **Traceable** farm-gate sourcing
- **Verified** moisture and quality certificates
- **Automated** invoicing

Agriculture Dept. →
- **Live data** district-level yield
- **Tracking** crop disease monitoring
- **DBT** subsidy disbursement

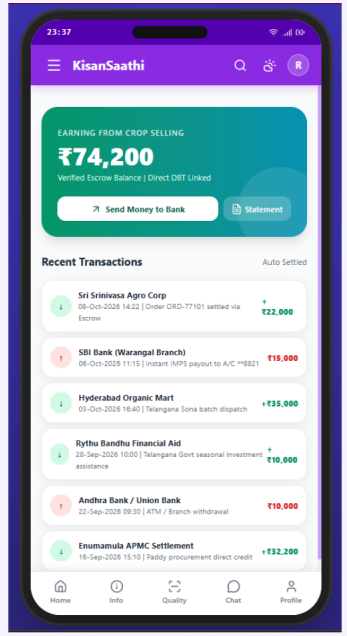
Economic
₹4,200–6,500 extra earnings per MT of Telangana Sona (RNR 15048)

Operational
Freight cut from ₹3.80/Kg to ₹2.35/Kg

Social
Inclusive access for Warangal, Medchal, Secunderabad, Nizamabad



Direct negotiation



Escrow wallet

Per-ton snapshot (illustrative)

- **Freight:** ₹3,800 → ₹2,350, saves ₹1,450 per MT
- **Commission:** 12–18% → 0%
- **Payment:** 15–45 days → 24 hours

TEAM FOREVERKNIGHTS

RESEARCH AND REFERENCES

HACK-X 2026
35-Hour Hackathon · MJCET

PROBLEM EVIDENCE

NITI Aayog

Agricultural Marketing and Farmer Friendly Reforms Index: intermediary margin losses

niti.gov.in

CROP DATA

Directorate of Economics & Statistics, Telangana

Major crop productivity reports: Telangana Sona and Cotton

des.telangana.gov.in

POLICY

Ministry of Agriculture & Farmers Welfare

Soil Health Card Scheme and PMFBY operational guidelines

soilhealth.dac.gov.in · pmfby.gov.in

AGRONOMY

ICAR-CRIDA & PJTSAU Rajendranagar

Soil nutrient thresholds and optimal agronomic guidelines

icar-crida.res.in · pjtsau.edu.in

RESEARCH

IEEE / ScienceDirect

Computer vision in grain quality inspection and defect estimation

ieeexplore.ieee.org · sciencedirect.com

OPEN-SOURCE TOOLS

OpenCV · TensorFlow.js · Leaflet.js

Grading engine, client-side AI and GPS route tracking

opencv.org · tensorflow.org/js · leafletjs.com

"Traditional portals merely list contact numbers. KisanSaathi standardizes quality with AI, removes freight barriers through clustering, and guarantees payment with escrow."